

Phase Box Firmware & Core Architecture

Handover Report & Current Engineering Status | RP2040 Video Synth Modulation Engine

1. System Overview & Architecture

Phase Box is a touring-ready hardware unit powered by the Raspberry Pi Pico (RP2040) functioning as an **algorithmic modulation engine** for external video synthesizers (such as LZX, Auto Waaaves, and `r_e_c_u_r`). The software architecture employs a strict Hardware Abstraction Layer (HAL) to decouple C++ core modulation logic from RP2040 hardware drivers, allowing full unit-testing and simulation on a host machine (macOS/Linux) via Google Test.

Performance critical timing is maintained via a dual-core architecture:

- **Core 0:** Real-time phase accumulators, LFO updating, tap tempo, MIDI clock processing, and UART MIDI CC output.
- **Core 1:** User interface event handling, OLED screen rendering, rotary encoder menus, and SPI Flash preset management.

2. Implemented Core Modules & Test Status

Module Header / Path	Description & Key Capabilities	Host Test Status
PhaseLFO include/core/phase_lfo.hpp	32-bit phase accumulator with 256-entry Sine LUT. Supports 5 waveforms: Sine, Triangle, Square, Sample & Hold, Turing Mutation (shift register). Adjustable mutation % & phase reset.	PASSED (Host gtest)
ClockManager include/core/clock_manager.hpp	Calculates BPM from footswitch delta times (20–300 BPM). Parses 24 PPQN MIDI clock bytes. Automatic fallback to internal tap after 1 sec timeout.	PASSED (Host gtest)
MidiRouter include/core/midi_router.hpp	Routes LFO outputs across 4 independent MIDI CC channels (1–16) and configurable CC numbers (0–127). Formats 3-byte MidiMessage structs.	PASSED (Host gtest)
ModulationEngine include/core/modulation_engine.hpp	Top-level core system aggregating 4 LFOs, ClockManager, and MidiRouter. Exposes tick(uint32_t current_time_us) returning std::vector<MidiMessage>.	PENDING FIX (Test Assert Index)

3. Immediate Bug Analysis & Fix for Next Session

Root Cause of Compilation Error in tests/test_main.cpp:

In ModulationEngineTest, assertions attempt to access struct properties directly on the vector container (msgs.status, msgs.data1) rather than indexing individual elements. Because msgs is a `std::vector<MidiMessage>`, the C++ compiler raises type mismatch errors.

Required Code Fix in tests/test_main.cpp:**Incorrect Line:**

```
EXPECT_EQ(msgs.status, 0xB0); // Error: no member named 'status' in vector
```

Correct Line (Indexed Access):

```
EXPECT_EQ(msgs[0].status, 0xB0); // Accesses first MidiMessage in vector  
EXPECT_EQ(msgs[0].data1, 14);    // CC #14 for LF0 0  
EXPECT_EQ(msgs[1].status, 0xB1); // Accesses second MidiMessage in vector  
EXPECT_EQ(msgs[1].data1, 15);    // CC #15 for LF0 1
```

4. Action Roadmap for Continuation

- **Step 1: Fix `tests/test\_main.cpp`** — Update vector access to `msgs[0]` and `msgs[1]` in `ModulationEngineTest`.`
- **Step 2: Run Host Tests** — Execute `cd build-host && make && ./tests/run_host_tests` to verify full green build.
- **Step 3: Core 0 Task Execution** — Implement real-time interrupt loop calling `engine.tick()` at 1000 Hz and driving RP2040 UART MIDI output.
- **Step 4: Core 1 GUI & Drivers** — Implement SSD1306 OLED driver, rotary encoder menu state machine, and preset saving to onboard SPI flash.